Motion, Speed, and Acceleration - Course Outline

Curriculum outline

Course context

Subject: Physics. Topic: Motion, Speed, and Acceleration. Level: beginner. Suggested duration: 75 minutes.

The lesson focuses on distance, displacement, speed, velocity. The remaining ideas are introduced after the core concepts are clear.

The lesson should move from a short concrete example into guided practice and then independent practice. Learners should explain at least one answer in their own words.

Learning objectives

- Explain the main idea behind distance.
- Explain the main idea behind displacement.
- Explain the main idea behind speed.
- Explain the main idea behind velocity.
- Apply acceleration in a short problem or example.
- Recognize a common mistake involving graphs.

Content boundaries

Keep the lesson centered on Motion, Speed, and Acceleration. Don't add advanced material that really belongs in a separate lesson.

New vocabulary should be defined once, then used consistently. If a formula, command, or process is introduced, include at least one worked example and one short practice item.

Include at least one misconception or near-miss that a beginner could realistically make.

Suggested lesson sequence

1. Opening

Start with a familiar situation or small problem connected to Motion, Speed, and Acceleration. Ask one diagnostic question before teaching formal terms.

Keep this section short. The aim is to surface what learners already think.

2. Core explanation

Introduce the main ideas in this order: distance, displacement, speed, velocity. Use one idea at a time and keep examples small enough to trace or calculate by hand.

Avoid jumping straight to the final rule without showing why the rule is useful.

3. Guided example

Work through one complete example and explain each step. The example should be similar to, but not identical to, later practice.

Ask learners to predict the next step before revealing it where practical.

4. Practice and check

Include a mix of recognition, prediction, correction, and short construction. A learner should not be able to pass by memorizing one definition.

End with a short check that matches the learning objectives.

Assessment notes

Use 5-8 short items for a 60-75 minute lesson. Mix question types. Include at least one item where a learner must explain why an answer is correct or identify why a tempting answer is wrong.

Avoid introducing new terminology inside the assessment unless the question supplies the meaning.